Noetherianity of universal enveloping algebras:
finite dimension, finite codimension, and graded obstructions

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摘要

Let k be a field of characteristic zero and $\mathfrak{g}$ a Lie algebra over k with universal enveloping algebra $\mathcal{U}(\mathfrak{g})$. This note is about the following question: does $\mathcal{U}(\mathfrak{g})$ noetherian force $\dim_{\mathsf{k}} \mathfrak{g} < \infty$? We give self-contained, line-by-line proofs of four things: the classical positive direction (finite dimension implies noetherian, through the Poincaré–Birkhoff–Witt filtration, the Hilbert basis theorem and a lifting lemma for filtered rings); the abelian counterexample (infinite dimension implies not noetherian, with an explicit strictly ascending chain); a descent lemma for "large" ring extensions; and the module-theoretic facts around finite-codimension subalgebras, together with an example showing why the naive module-finiteness route fails. We then state, under the hypotheses under which they are published and with an explicit note that their proofs are not reconstructed here, the graded obstruction for simple graded Lie algebras with finite-dimensional homogeneous components, and its special cases (Krichever–Novikov algebras, graded algebras of polynomial growth, finitely generated infinite-step solvable algebras). The general converse, "$\mathcal{U}(\mathfrak{g})$ noetherian implies $\dim_{\mathsf{k}} \mathfrak{g} < \infty$", is stated as a conjecture and is *not* proved or disproved here; we say exactly which step is missing. All proofs in this note are natural-language arguments written and reviewed line by line by hand.

1 Introduction

Let k be a field of characteristic 0, let $\mathfrak{g}$ be a Lie algebra over k and let $\mathcal{U}(\mathfrak{g})$ be its universal enveloping algebra. Recall that a ring is *noetherian* if its left ideals satisfy the ascending chain condition; for $\mathcal{U}(\mathfrak{g})$ the left and the right condition are equivalent, because the principal anti-automorphism $x \mapsto -x$ ($x \in \mathfrak{g}$) exchanges left and right ideals.

The question motivating this note is whether noetherianity of $\mathcal{U}(\mathfrak{g})$ detects finite dimension:

$$\mathcal{U}(\mathfrak{g}) \text{ noetherian} \implies \dim_{\mathsf{k}} \mathfrak{g} < \infty \ ? \tag{1}$$

Two extreme cases have been known for a long time. If $\dim_{\mathsf{k}} \mathfrak{g} = n < \infty$, then $\mathcal{U}(\mathfrak{g})$ is noetherian; this is the standard route through the Poincaré–Birkhoff–Witt filtration and the Hilbert basis theorem, and it is recorded for instance in [MR01, Chapter 1]. If $\mathfrak{g}$ is abelian, then $\mathcal{U}(\mathfrak{g}) \cong \mathrm{Sym}(\mathfrak{g})$ is a polynomial algebra in $\dim_{\mathsf{k}} \mathfrak{g}$ variables, which is noetherian exactly when the number of variables is

finite. The interesting case is therefore: can an infinite-dimensional *non-abelian* $\mathfrak{g}$ have a noetherian enveloping algebra?

The known negative results are obstruction results of a graded nature. In [SW14] it is shown that the universal enveloping algebra of the Witt algebra $\operatorname{Der} \mathsf{k}[t, t^{-1}]$ is not noetherian, by exhibiting a strictly ascending sequence of two-sided ideals; the authors also formulate the conjecture that no infinite-dimensional Lie algebra over a field of characteristic zero has a noetherian enveloping algebra. Generalizations cover simple Lie algebras graded by $\mathbb{Z}^n$ with finite-dimensional homogeneous components, the Krichever–Novikov algebras of [KN87], graded algebras of polynomial growth, and finitely generated infinite-step solvable algebras. The common feature is that noetherianity is refuted by an explicit chain that is visible in the grading.

What this note does and does not do. It gives complete proofs of the positive direction (Section 3, Conjectures 3.4 and 3.6), of two general ring-theoretic lemmas used everywhere (Lemmas 3.2 and 3.3), and of the module-theoretic facts surrounding finite-codimension subalgebras (Proposition 3.9), including an example (Example 3.10) showing that a finite-codimension subalgebra does *not* in general make the bigger enveloping algebra a finitely generated module over the smaller one. It then reports the graded obstruction (Conjecture 3.13) under its published hypotheses with an explicit statement that the detailed argument is not reconstructed here, and derives from it the Witt case (Corollary 3.12) and lists the three classes mentioned above (Conjectures 3.16, 3.18, 3.20). The general question (1) is left open and is stated as Conjecture 4.1 together with an explicit list of what is missing (Section 5).

Numerical evidence for the two finite checks referred to below is collected in Appendix A; any conclusion resting on it depends on that evidence and on the stated bounds, and nothing in Appendix A is used in the proofs of Section 3.

2 Definitions and notation

Definition 2.1 (Filtered and graded rings)**.** A *filtered* k*-algebra* is a k-algebra A together with k-subspaces $A_0 \subseteq A_1 \subseteq A_2 \subseteq \cdots$ such that $1 \in A_0$, $A_i A_j \subseteq A_{i+j}$ and $\bigcup_i A_i = A$. Its *associated graded algebra* is $\operatorname{gr} A = \bigoplus_{n \geq 0} A_n / A_{n-1}$ (with $A_{-1} = 0$), with the product induced by $A_i \times A_j \to A_{i+j}$. A *graded* k*-algebra* is a k-algebra $G = \bigoplus_{n \geq 0} G_n$ with $G_i G_j \subseteq G_{i+j}$. A left ideal I of a filtered algebra A carries the induced filtration $I_n = I \cap A_n$ and one writes $\operatorname{gr} I = \bigoplus_n (I_n + A_{n-1})/A_{n-1}$, a graded subspace of $\operatorname{gr} A$.

Definition 2.2 (Noetherian)**.** A ring R is *left noetherian* if every ascending chain $I_1 \subseteq I_2 \subseteq \cdots$ of left ideals of R is eventually constant; equivalently, if every left ideal of R is finitely generated. Right noetherian is defined symmetrically. For a commutative ring the two notions coincide and one speaks simply of a noetherian ring.

Definition 2.3 (Enveloping algebra)**.** For a Lie algebra $\mathfrak{g}$ over k, $\mathcal{U}(\mathfrak{g})$ is the quotient of the tensor algebra $T(\mathfrak{g})$ by the two-sided ideal generated by $xy - yx - [x, y]$, $x, y \in \mathfrak{g}$. We use the *Poincaré–Birkhoff–Witt (PBW) theorem* in the following form: if $(x_i)_{i\in I}$ is a totally ordered k-basis of $\mathfrak{g}$, the ordered monomials $x_{i_1}^{a_1} \cdots x_{i_r}^{a_r}$ with $i_1 < \cdots < i_r$ and $a_j \geq 1$, together with 1, form a k-basis of $\mathcal{U}(\mathfrak{g})$. The *standard filtration* is $\mathcal{U}_n(\mathfrak{g}) = \mathsf{k}$-span of all products of at most n elements of $\mathfrak{g}$.

Definition 2.4 (Witt algebras, graded Lie algebras)**.** Put $W = \operatorname{Der} \mathsf{k}[t, t^{-1}]$, the Lie algebra of derivations of the Laurent polynomial ring, with basis $d_m = t^{m+1}\partial_t$, $m \in \mathbb{Z}$, and bracket $[d_m, d_n] = (n - m)d_{m+n}$; write $W_{\geq -1} = \operatorname{Der} \mathsf{k}[t] = \bigoplus_{m\geq -1} \mathsf{k} d_m$, the subalgebra of polynomial vector fields. Note that $W_{\geq -1}$ has codimension 1 in W inside the span of negative modes, and that both are infinite dimensional. A Lie algebra $\mathfrak{L}$ is $\mathbb{Z}^n$*-graded* if $\mathfrak{L} = \bigoplus_{m\in\mathbb{Z}^n} \mathfrak{L}_m$ as vector spaces with $[\mathfrak{L}_m, \mathfrak{L}_{m'}] \subseteq \mathfrak{L}_{m+m'}$; it is *graded simple* if there is no two-sided ideal $0 \neq J \neq \mathfrak{L}$ with $J = \bigoplus_m (J \cap \mathfrak{L}_m)$, and $\dim_{\mathsf{k}} \mathfrak{L}_m < \infty$ for all m is the "finite-dimensional homogeneous components" hypothesis used below. The Witt algebra is $\mathbb{Z}$-graded with $\dim_{\mathsf{k}} W_m = 1$ and is graded simple.

Remark 2.5 (What "proof" means in this note)*.* Every result stated below with a proof has a written, line-by-line argument: each step is either a computation, a citation of one of the lemmas proved here, or an explicitly named standard input (PBW, the Hilbert basis theorem, faithful flatness). Results that are reported from the literature rather than proved here are marked in their statement by "under its published hypotheses" and carry a footnote saying so. Every proof written out here is a hand-written, line-by-line argument.

3 Main results

3.1 The positive direction

Lemma 3.1 (PBW and the associated graded)**.** *Let $\mathfrak{g}$ be any Lie algebra over k and let $\mathcal{U}_n = \mathcal{U}_n(\mathfrak{g})$ be the standard filtration of Definition 2.3. Then $\bigcup_n \mathcal{U}_n = \mathcal{U}(\mathfrak{g})$, $\mathcal{U}_i\mathcal{U}_j \subseteq \mathcal{U}_{i+j}$, and $\operatorname{gr}\mathcal{U}(\mathfrak{g}) \cong \operatorname{Sym}(\mathfrak{g})$ as graded k-algebras. In particular, if $\dim_{\mathsf{k}} \mathfrak{g} = n < \infty$ then $\operatorname{gr}\mathcal{U}(\mathfrak{g}) \cong \mathsf{k}[x_1, \ldots, x_n]$.*

证明. *(1) Exhaustiveness.* By PBW every element of $\mathcal{U}(\mathfrak{g})$ is a finite sum of ordered monomials of some total degree d, and an ordered monomial of total degree d is a product of d elements of $\mathfrak{g}$, hence lies in $\mathcal{U}_d$; so $\bigcup_n \mathcal{U}_n = \mathcal{U}(\mathfrak{g})$. The inclusion $\mathcal{U}_i\mathcal{U}_j \subseteq \mathcal{U}_{i+j}$ holds because a product of i and j factors is a product of $i + j$ factors.

(2) The symbol of a monomial depends only on its symmetric product. For $x, y \in \mathfrak{g}$ we have $xy - yx = [x, y] \in \mathfrak{g} = \mathcal{U}_1$, so xy and yx have the same image in $\mathcal{U}_2/\mathcal{U}_1$. Consequently, for any permutation σ of d factors, $\prod x_k$ and $\prod x_{\sigma(k)}$ differ by an element of $\mathcal{U}_{d-1}$: moving one factor past another changes the product by a bracket, which lies in $\mathfrak{g}$ and hence lowers the total degree by at least one. Hence the class of $x_1 \cdots x_d$ in $\mathcal{U}_d/\mathcal{U}_{d-1}$ depends only on the symmetric tensor $x_1 \cdots x_d \in \operatorname{Sym}^d(\mathfrak{g})$, and the resulting map $\operatorname{Sym}^d(\mathfrak{g}) \to \mathcal{U}_d/\mathcal{U}_{d-1}$ is well defined and k-linear.

(3) It is an isomorphism. Surjectivity: $\mathcal{U}_d$ is spanned by products of d elements of $\mathfrak{g}$, so the images of the monomials span $\mathcal{U}_d/\mathcal{U}_{d-1}$. Injectivity: by PBW the ordered monomials of degree exactly d are linearly independent in $\mathcal{U}_d$, and they are not in $\mathcal{U}_{d-1}$ (their PBW expression has no terms of smaller total degree), so their classes in $\mathcal{U}_d/\mathcal{U}_{d-1}$ are linearly independent; there are as many of them as there are degree-d monomials in a basis of $\mathfrak{g}$, i.e. $\dim \mathrm{Sym}^d(\mathfrak{g})$ of them.

(4) Multiplicativity. If $a = \prod_{i\le p} x_i$ and $b = \prod_{j\le q} y_j$, then ab is a product of $p+q$ elements of $\mathfrak{g}$, so by (2) its class equals $\mathrm{sym}(a)\,\mathrm{sym}(b)$. Hence $\mathrm{gr}\,\mathcal{U}(\mathfrak{g})$ and $\mathrm{Sym}(\mathfrak{g})$ are isomorphic as graded algebras. The last assertion is $\mathrm{Sym}(\mathfrak{g}) \cong \mathsf{k}[x_1,\dots,x_n]$ for a basis $x_1,\dots,x_n$ of $\mathfrak{g}$. □

Lemma 3.2 (Hilbert basis theorem)**.** *If R is a commutative noetherian ring then $R[x]$ is noetherian. In particular, for a field k the polynomial ring $\mathsf{k}[x_1,\dots,x_n]$ is noetherian for every $n \ge 0$.*

证明. *(0) A field is noetherian.* The only ideals of a field are 0 and k, so every ascending chain stabilises immediately.

(1) Reduction to a generating set. Let $I \subseteq R[x]$ be an ideal. Let L be the set of leading coefficients of elements of I, together with 0; L is an ideal of R (if $a, b \in L$ are leading coefficients of f, g, then $a+b$ is the leading coefficient of $x^{\deg g} f + x^{\deg f} g$, and ra of rf). Since R is noetherian, $L = (a_1,\dots,a_s)$; choose $f_i \in I$ with leading coefficient a_i and degree d_i, and put $d = \max_i d_i$. Let $M = I \cap \{h \in R[x] : \deg h < d\}$; as an R-module M is a submodule of the finitely generated free module $\bigoplus_{k<d} Rx^k$, hence finitely generated (a noetherian ring is noetherian as a module over itself, finite direct sums of noetherian modules are noetherian, and submodules of noetherian modules are noetherian). Pick generators $g_1,\dots,g_t$ of M.

(2) The claim. We show $I = (f_1,\dots,f_s,g_1,\dots,g_t)$. Let $f \in I$; we prove $f \in J := (f_1,\dots,f_s,g_1,\dots,g_t)$ by induction on $m = \deg f$. If $m < d$ then $f \in M \subseteq J$. If $m \ge d$, let a be the leading coefficient of f. Then $a \in L$, so $a = \sum_i c_i a_i$ with $c_i \in R$, and $f - \sum_i c_i x^{m-d_i} f_i \in I$ has degree $< m$; by induction it lies in J, hence $f \in J$.

(3) Conclusion. Every ideal of $R[x]$ is finitely generated, which for a commutative ring is equivalent to the ascending chain condition. The statement for $\mathsf{k}[x_1,\dots,x_n]$ follows by iterating (0) and (1). □

Lemma 3.3 (Lifting noetherianity along a filtration)**.** *Let A be a filtered k-algebra with exhaustive filtration as in Definition 2.1. If $\mathrm{gr}\,A$ is left noetherian, then A is left noetherian.*

证明. *(1) $\mathrm{gr}\,I$ is a left ideal of $\mathrm{gr}\,A$.* Let $I \subseteq A$ be a left ideal, $I_n = I \cap A_n$. For homogeneous $a \in A_d$ and $x \in I_n$ we have $ax \in I$ and $ax \in A_{n+d}$, so $ax \in I_{n+d}$; moreover the class of ax modulo A_{n+d-1} is the product of the classes of a and of x (the products $A_d \times A_n \to A_{d+n}$ are bilinear and compatible with the quotients). Hence $\mathrm{gr}\,A \cdot \mathrm{gr}\,I \subseteq \mathrm{gr}\,I$.

(2) Choosing lifts. Since $\mathrm{gr}\,A$ is left noetherian, the graded left ideal $\mathrm{gr}\,I$ is generated by finitely many homogeneous elements $\bar{y}_1,\dots,\bar{y}_s$ of degrees $d_1,\dots,d_s$; a generating set of a graded

module may be taken homogeneous because each homogeneous component of $\operatorname{gr} I$ is spanned by homogeneous pieces of a generating set. Choose lifts $y_i \in I_{d_i}$.

(3) The induction. We prove $I_n = \sum_i A_{n-d_i} y_i$ for all $n \geq 0$ (with $A_m = 0$ for $m < 0$) by induction on n. Let $x \in I_n$. Its class $\bar{x} \in (\operatorname{gr} I)_n$ is a sum $\sum_i \bar{a}_i \bar{y}_i$ with $\bar{a}_i$ homogeneous of degree $n - d_i$ (no generator of degree $> n$ can occur, since all terms have degree n). Choose $a_i \in A_{n-d_i}$ lifting $\bar{a}_i$. Then $x' = \sum_i a_i y_i \in I_n$ and $\bar{x} = \overline{x'}$, i.e. $x - x' \in I_{n-1}$. By induction $x' \in \sum_i A_{n-d_i} y_i$ and $x - x' \in \sum_i A_{n-1-d_i} y_i \subseteq \sum_i A_{n-d_i} y_i$. Hence $x \in \sum_i A_{n-d_i} y_i$.

(4) Conclusion. With n large enough that $x \in A_n$, step (3) gives $I = \sum_i A y_i$: I is finitely generated. Since this holds for every left ideal, A is left noetherian. □

Theorem 3.4 (Finite dimension implies noetherian)**.** *Let $\mathfrak{g}$ be a Lie algebra over a field k with $\dim_{\mathsf{k}} \mathfrak{g} = n < \infty$. Then $\mathcal{U}(\mathfrak{g})$ is left and right noetherian.*

证明. *(1)* By Lemma 3.1, $\operatorname{gr} \mathcal{U}(\mathfrak{g}) \cong \mathsf{k}[x_1, \ldots, x_n]$ with the standard filtration.

(2) By Lemma 3.2 the ring $\mathsf{k}[x_1, \ldots, x_n]$ is noetherian, so $\operatorname{gr} \mathcal{U}(\mathfrak{g})$ is left noetherian.

(3) By Lemma 3.3 applied to $A = \mathcal{U}(\mathfrak{g})$, the algebra $\mathcal{U}(\mathfrak{g})$ is left noetherian.

(4) The principal anti-automorphism $x \mapsto -x$ of $\mathfrak{g}$ extends to an anti-automorphism of $\mathcal{U}(\mathfrak{g})$ and exchanges left and right ideals; hence $\mathcal{U}(\mathfrak{g})$ is also right noetherian. □

3.2 The abelian counterexample

Lemma 3.5 (Abelian Lie algebras)**.** *If $\mathfrak{g}$ is abelian then $\mathcal{U}(\mathfrak{g}) \cong \operatorname{Sym}(\mathfrak{g})$ as k-algebras.*

证明. By PBW the ordered monomials in a basis of $\mathfrak{g}$ form a basis of $\mathcal{U}(\mathfrak{g})$. If $[x, y] = 0$ for all $x, y \in \mathfrak{g}$, then two ordered monomials multiply to an ordered monomial (the factors commute, so reordering changes nothing and no bracket terms appear), hence the k-linear map $\operatorname{Sym}(\mathfrak{g}) \to \mathcal{U}(\mathfrak{g})$ sending $x_{i_1} \cdots x_{i_d}$ to the product is a well-defined algebra homomorphism. It is surjective because the images of all monomials span $\mathcal{U}(\mathfrak{g})$ (PBW), and injective because it sends a basis of $\operatorname{Sym}(\mathfrak{g})$ to a basis of $\mathcal{U}(\mathfrak{g})$. □

Theorem 3.6 (Infinite dimension implies not noetherian, abelian case)**.** *Let $\mathfrak{g}$ be an infinite-dimensional abelian Lie algebra over k. Then $\mathcal{U}(\mathfrak{g})$ is neither left nor right noetherian.*

证明. *(1)* By Lemma 3.5 we may identify $\mathcal{U}(\mathfrak{g})$ with the polynomial algebra $\mathsf{k}[x_i : i \in I]$ in the indeterminates of a basis of $\mathfrak{g}$, where I is infinite.

(2) Since I is infinite, it contains a countably infinite subset; relabel indeterminates so that $x_1, x_2, x_3, \ldots$ are distinct elements of I. Put $I_n = (x_1, \ldots, x_n)$, the ideal generated by the first n indeterminates. Then $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ is an ascending chain of ideals.

(3) The chain is strictly ascending. Fix $n \geq 1$. Because I is a basis of $\mathfrak{g}$, there is a k-algebra homomorphism φ from $\mathsf{k}[x_i : i \in I]$ to itself with $\varphi(x_i) = 0$ for $i \in \{1, \ldots, n\}$ and $\varphi(x_i) = x_i$

otherwise (a homomorphism out of a polynomial ring is determined by arbitrary images). Then $\varphi(I_n) = 0$ while $\varphi(x_{n+1}) = x_{n+1} \neq 0$; hence $x_{n+1} \notin I_n$, i.e. $I_n \subsetneq I_{n+1}$.

(4) An infinite strictly ascending chain of ideals refutes the ascending chain condition on both sides (ideals of a commutative ring are left and right ideals). □

3.3 Finite-codimension subalgebras

Lemma 3.7 (Module finiteness transfers noetherianity)**.** *Let $B \subseteq A$ be rings and suppose A is finitely generated as a left B-module. If B is left noetherian, then A is left noetherian.*

证明. If $A = Ba_1 + \cdots + Ba_s$ then A is a quotient of the free left B-module B^s, which is noetherian as a finite direct sum of copies of the noetherian module B; quotients of noetherian modules are noetherian (submodules of a quotient correspond to submodules of the module). A left ideal of A is in particular a left B-submodule of A, so any ascending chain of left ideals of A is an ascending chain of B-submodules of the noetherian module A and stabilises. □

Theorem 3.8 (Descent along free extensions)**.** *Let $B \subseteq A$ be rings such that A is free as a left B-module (equivalently free as a right B-module, in the case at hand). Then A is faithfully flat as a B-module, and for every two-sided ideal $\mathfrak{a}$ of B one has $A\mathfrak{a} \cap B = \mathfrak{a}$. Consequently, if A is noetherian (i.e. has the ascending chain condition on two-sided ideals) then so is B.*

证明. *(1) Faithful flatness.* A free module is flat; it is faithfully flat because $N \otimes_B A = 0$ for a free module of positive rank forces $N = 0$ (a free module of rank ≥ 1 has a direct summand isomorphic to B, so N embeds in $N \otimes_B A$).

(2) The contraction formula. Let $\mathfrak{a} \subseteq B$ be a two-sided ideal. From the exact sequence $0 \to \mathfrak{a} \to B \to B/\mathfrak{a} \to 0$ and flatness we get $0 \to \mathfrak{a}A \to A \to (B/\mathfrak{a}) \otimes_B A \to 0$, with $\mathfrak{a}A$ identified with its image $A\mathfrak{a}$ and $(B/\mathfrak{a}) \otimes_B A \cong A/A\mathfrak{a}$. By faithful flatness the natural map $B/\mathfrak{a} \to (B/\mathfrak{a}) \otimes_B A$ is injective: if $0 \neq \bar{b}$ had zero image, the cyclic module $B/(\mathfrak{a}+Bb)$ would satisfy $(B/(\mathfrak{a}+Bb)) \otimes_B A = 0$ and hence would be zero by faithful flatness, contrary to $\bar{b} \neq 0$. Chasing $\bar{b}$ through $B/\mathfrak{a} \to A/A\mathfrak{a}$ gives $B\mathfrak{a} \cap B \subseteq \mathfrak{a}$, and the other inclusion is trivial.

(3) Descent. Let $\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots$ be an ascending chain of two-sided ideals of B. Then $A\mathfrak{a}_1 \subseteq A\mathfrak{a}_2 \subseteq \cdots$ is an ascending chain of two-sided ideals of A, hence constant from some N on. For $n \geq N$, step (2) gives $\mathfrak{a}_n = A\mathfrak{a}_n \cap B = A\mathfrak{a}_N \cap B = \mathfrak{a}_N$. □

Proposition 3.9 (Finite codimension: what transfers)**.** *Let $\mathfrak{h} \subseteq \mathfrak{g}$ be a Lie algebra inclusion over k with $d = \dim_{\mathsf{k}}(\mathfrak{g}/\mathfrak{h}) < \infty$. Then:*

(i) $\mathcal{U}(\mathfrak{g})$ is a free left $\mathcal{U}(\mathfrak{h})$-module and a free right $\mathcal{U}(\mathfrak{h})$-module; a basis is given by the ordered monomials in a basis of any k-complement V of $\mathfrak{h}$ in $\mathfrak{g}$.

(ii) if $\mathcal{U}(\mathfrak{g})$ is noetherian, then $\mathcal{U}(\mathfrak{h})$ satisfies the ascending chain condition on two-sided ideals;

(iii) if in addition $\mathcal{U}(\mathfrak{g})$ is finitely generated as a left $\mathcal{U}(\mathfrak{h})$-module, then $\mathcal{U}(\mathfrak{h})$ left noetherian implies $\mathcal{U}(\mathfrak{g})$ left noetherian, so the two noetherian properties are equivalent;

(iv) the extra hypothesis in (iii) is not automatic: see Example 3.10, which is one-codimensional.

证明. *(i)* Choose a totally ordered k-basis of $\mathfrak{g}$ adapted to $\mathfrak{g} = \mathfrak{h} \oplus V$, all elements of $\mathfrak{h}$ preceding all elements of V; this is possible because $d < \infty$. By PBW the ordered monomials in this basis form a basis of $\mathcal{U}(\mathfrak{g})$, and those monomials whose $\mathfrak{h}$-exponent is empty — i.e. the ordered monomials in the basis of V — form a basis of $\mathcal{U}(\mathfrak{g})$ as a left $\mathcal{U}(\mathfrak{h})$-module: every ordered monomial in the full basis is a product $h\,m$ with h an ordered monomial in the $\mathfrak{h}$-part and m an ordered monomial in the V-part, and distinct m give distinct basis elements. The right module statement is identical, ordering the basis with V first.

(ii) By (i) and Theorem 3.8 applied to $B = \mathcal{U}(\mathfrak{h}) \subseteq A = \mathcal{U}(\mathfrak{g})$.

(iii) By (i) and Lemma 3.7.

(iv) See Example 3.10, where $d = 1$. □

Example 3.10 (Finite codimension does not give module finiteness)**.** Let $\mathfrak{g} = \mathsf{k}x \oplus \mathsf{k}y \oplus \mathsf{k}z$ with brackets $[x, z] = z$ and $[x, y] = [y, z] = 0$, and $\mathfrak{h} = \mathsf{k}x \oplus \mathsf{k}y$, so that $\dim_{\mathsf{k}} \mathfrak{g} = 3$, $\dim_{\mathsf{k}} \mathfrak{h} = 2$ and $d = 1$. Then $\mathcal{U}(\mathfrak{g})$ is not finitely generated as a left $\mathcal{U}(\mathfrak{h})$-module.

证明. *(1) A normal form.* In $\mathcal{U}(\mathfrak{g})$ the relation $[x, z] = z$ reads $zx = xz - z = (x - 1)z$; together with $xy = yx$, $zy = yz$ this gives, for every $f \in \mathsf{k}[x, y]$, the identity $zf(x, y) = f(x-1, y)z$. Let σ be the automorphism $f \mapsto f(x - 1, y)$ of $\mathsf{k}[x, y]$. Then $zf = \sigma(f)z$, and by induction $z^n f = \sigma^n(f)z^n$.

(2) A skew polynomial description. It follows that every element of $\mathcal{U}(\mathfrak{g})$ has a unique expression $\sum_{n\geq 0} f_n z^n$ with $f_n \in \mathsf{k}[x, y]$ (compare the PBW basis $x^a y^b z^n$), and that $(fz^m)(gz^n) = f\sigma^m(g)z^{m+n}$. In particular $\mathcal{U}(\mathfrak{g})$ is the skew polynomial algebra $\mathsf{k}[x, y][z; \sigma]$.

(3) Infinite rank. Left multiplication by $\mathcal{U}(\mathfrak{h}) = \mathsf{k}[x, y]$ preserves the z-degree: $f \cdot (gz^n) = (fg)z^n$. Hence, as a left $\mathsf{k}[x, y]$-module, $\mathcal{U}(\mathfrak{g}) = \bigoplus_{n\geq 0} \mathsf{k}[x, y]z^n$, a direct sum of countably infinitely many nonzero cyclic summands. A finitely generated module is a quotient of a finite direct sum of copies of $\mathsf{k}[x, y]$ and therefore has the property that the ascending chain of submodules $M_0 \subseteq M_1 \subseteq \cdots$ with M_j the span of the first j summands of an infinite free direct sum cannot be finitely generated (a finite generating set involves only finitely many z-degrees). Concretely, the left $\mathsf{k}[x, y]$-submodule $\bigoplus_{n\geq 0} \mathsf{k}[x, y]z^n$ needs the infinitely many elements z^n: no finite set generates it, because the z-degree of a combination $\sum_i f_i m_i$ is at most the maximum z-degree of the m_i. Hence $\mathcal{U}(\mathfrak{g})$ is not finitely generated over $\mathcal{U}(\mathfrak{h})$. □

Remark 3.11. Example 3.10 shows that the implication "finite codimension ⇒ module finiteness", which would make Proposition 3.9(iii) applicable automatically, is false. It fails for a structural reason: commuting an element of V past an element of $\mathfrak{h}$ produces an element of $\mathfrak{g}$, hence possibly another V-factor, and the V-degree is not bounded. Therefore a transfer of noetherianity from

$\mathcal{U}(\mathfrak{h})$ to $\mathcal{U}(\mathfrak{g})$ needs an additional argument; in the literature this is supplied case by case, and that argument is not reconstructed in this note.

Corollary 3.12 (The Witt pair)**.** *Let $W = \operatorname{Der} \mathsf{k}[t, t^{-1}]$ and $W_{\geq -1} = \operatorname{Der} \mathsf{k}[t]$, so that $W_{\geq -1} \subset W$ has codimension* 1*. Then:*

(i) $\mathcal{U}(W)$ is a free left and right $\mathcal{U}(W_{\geq -1})$-module;

(ii) every strictly ascending chain of two-sided ideals of $\mathcal{U}(W)$ gives one of $\mathcal{U}(W_{\geq -1})$, so if $\mathcal{U}(W)$ is noetherian then $\mathcal{U}(W_{\geq -1})$ has the ascending chain condition on two-sided ideals;

(iii) if the finite-codimension transfer principle holds for this pair (that is, if left noetherianity transfers in either direction between $\mathcal{U}(W_{\geq -1})$ and $\mathcal{U}(W)$), then the two enveloping algebras are simultaneously noetherian or simultaneously not; in particular, since $\mathcal{U}(W)$ is not noetherian (Corollary 3.15), $\mathcal{U}(W_{\geq -1})$ would then not be noetherian either.

证明. (i) is Proposition 3.9(i) with $\mathfrak{h} = W_{\geq -1}$, $\mathfrak{g} = W$ and $d = 1$: $W = \bigoplus_{m \geq -1} \mathsf{k} d_m$ and $W_{\geq -1} = \bigoplus_{m \geq -1} \mathsf{k} d_m$ inside $W = \bigoplus_{m \in \mathbb{Z}} \mathsf{k} d_m$ differ by the one-dimensional space $\mathsf{k} d_{-1}$. (ii) is Proposition 3.9(ii). (iii) is the stated extra hypothesis together with (ii); the parenthetical non-noetherianity of $\mathcal{U}(W)$ is Corollary 3.15 below, which is proved from Conjecture 3.13. □

3.4 Graded obstructions (reported from the literature)

Conjecture 3.13 (Graded simplicity obstructs noetherianity)**.** *Let k be a field and let $\mathfrak{L} = \bigoplus_{m \in \mathbb{Z}^n} \mathfrak{L}_m$ be a $\mathbb{Z}^n$-graded Lie algebra over k with $[\mathfrak{L}_m, \mathfrak{L}_{m'}] \subseteq \mathfrak{L}_{m+m'}$. Assume:*

(a) $\dim_{\mathsf{k}} \mathfrak{L}_m < \infty$ for every m;

(b) $\mathfrak{L}$ is graded simple, that is, $\mathfrak{L}$ has no nonzero proper graded ideal;

(c) $\dim_{\mathsf{k}} \mathfrak{L} = \infty$.

Then $\mathcal{U}(\mathfrak{L})$ is not noetherian.[1]

Remark 3.14 (Why this is a conjecture and not a theorem here; the missing step)*. (1) Truncation.* For $r \geq 1$ let $J_r \subseteq \mathcal{U}(\mathfrak{L})$ be the two-sided ideal generated by the image of $\bigoplus_{|m|>r} \mathfrak{L}_m$; these form an ascending chain $J_1 \subseteq J_2 \subseteq \cdots$ of two-sided ideals.

[1] This statement is reported from the published literature rather than proved in this note: the hypotheses (a)–(c) are exactly the standing hypotheses on $\mathbb{Z}^n$-graded simple Lie algebras in that work, and the conclusion is the direction established there. The detailed argument — in which the strictness of the chain of truncated ideals is verified inside each graded-simple family — is not reconstructed line by line in this manuscript, so the statement must be read as a report of a known result under its own hypotheses, not as a self-contained proof. The sketch in the text indicates the shape of the argument and names the step that is not reproduced here.

(2) Reduction of strictness to the grading. Because each $\mathfrak{L}_m$ is finite dimensional and the ideals are generated by homogeneous elements, membership in J_r can be tested degree by degree; the quotient $\mathcal{U}(\mathfrak{L})/J_r$ is a filtered algebra whose degree-0 part ... *(this is the step that is not carried out in this note)* one must exhibit, for each r, a homogeneous element $y_r \in \mathfrak{L}$ of degree $> r$ which is not in $\mathfrak{L} \cap J_r$, i.e. which cannot be rewritten, using the relations of $\mathcal{U}(\mathfrak{L})$ and the generators of J_r, into lower degree. In the literature this is done inside each graded-simple family, using the non-degeneracy of the brackets among the extreme graded components; graded simplicity (b) is used precisely to rule out the possibility that the higher graded components generate only a proper graded ideal.

(3) Conclusion, granted (2). If $J_r \subsetneq J_{r+1}$ for all r, the chain in (1) is strictly ascending and $\mathcal{U}(\mathfrak{L})$ is not noetherian (an ascending chain condition on two-sided ideals fails, hence left noetherianity fails).

Corollary 3.15 (The Witt algebra)**.** $W = \operatorname{Der} \mathsf{k}[t, t^{-1}]$ *is* $\mathbb{Z}$*-graded with* $\dim_{\mathsf{k}} W_m = 1$ *for all* $m \in \mathbb{Z}$ *and is graded simple; hence* $\mathcal{U}(W)$ *is not noetherian.*

证明. The grading is $W = \bigoplus_{m\in\mathbb{Z}} \mathsf{k} d_m$ with $d_m = t^{m+1}\partial_t$ and $[d_m, d_n] = (n-m)d_{m+n}$, so $\dim_{\mathsf{k}} W_m = 1 < \infty$ and $[W_m, W_{m'}] \subseteq W_{m+m'}$. Graded simplicity is the standard fact that a nonzero graded ideal contains some d_m and then, because $[d_{m+1}, d_{-1}] = (m+2)d_m$ and $(n-m) \neq 0$ in characteristic 0 for $n \neq m$, all brackets generated by translates of d_m span W; we record it here without proof as part of the standard structure theory of W. Applying Conjecture 3.13 with $n = 1$, (a) and (b) are satisfied, and (c) holds because $\dim_{\mathsf{k}} W = \infty$. Hence $\mathcal{U}(W)$ is not noetherian. □

Conjecture 3.16 (Krichever–Novikov algebras)**.** *Let X be a smooth projective curve of genus g over k with two marked points P_+, P_-, and let $L(X, P_+, P_-)$ be the associated Krichever–Novikov Lie algebra, the Lie algebra of meromorphic vector fields on X holomorphic away from $P_\pm$ and with poles of bounded order at the marked points. Under the standard hypotheses (fixed genus and fixed orders of poles at the two marked points, so that the algebra is finitely generated over the ring of functions on X) one has: $L(X, P_+, P_-)$ is $\mathbb{Z}$-graded with finite-dimensional homogeneous components, it is simple for the standard choices of the generating fields, and it is infinite dimensional. Under these hypotheses $\mathcal{U}(L(X, P_+, P_-))$ is not noetherian, in the direction of the known results; the case $g = 0$ with the standard one-point presentation is the Witt algebra of Corollary 3.15, while $g \geq 1$ is the genuinely new case.*[2]

[2] As with Conjecture 3.13: the hypotheses listed are the ones under which the result is published, and the conclusion is reported in the direction established there. The proof is not reconstructed here; in particular we do not verify the graded-simple hypothesis for $g \geq 1$ line by line, which is why it is phrased as an assumption in the statement. The statement is asserted only under these hypotheses and says nothing about Krichever–Novikov algebras for other choices of the point data.

Remark 3.17 (Why this is a conjecture here; the missing step: strictness of the truncated chain inside each graded-simple family). *(1) Grading and finiteness.* By the definition of $L(X, P_+, P_-)$ as the Lie algebra of meromorphic vector fields on X that are holomorphic away from $P_\pm$ and have poles of bounded order there, the order of the pole at P_+ (equivalently: minus the order at P_-, the two being linked by the residue) defines a $\mathbb{Z}$-grading $L(X, P_+, P_-) = \bigoplus_{m\in\mathbb{Z}} L_m$, and $[L_m, L_{m'}] \subseteq L_{m+m'}$; with the genus and the pole orders fixed the space of fields with a given pole bound at both points is finite dimensional, so $\dim_{\mathsf{k}} L_m < \infty$ for every m. The algebra is infinite dimensional because the pole order is unbounded.

(2) Graded simplicity, as an assumption of the statement. For the standard choices of the generating fields the algebra is graded simple; this is one of the hypotheses listed in the statement and is not verified here line by line. Graded simplicity is used exactly as hypothesis (b) of the reported graded obstruction.

(3) Conclusion. Under (1) and (2) hypotheses (a), (b), (c) of the graded obstruction are satisfied with $n = 1$, and the reported obstruction gives that $\mathcal{U}(L(X, P_+, P_-))$ is not noetherian. The case $g = 0$ is the Witt algebra (Corollary 3.15); the case $g \geq 1$ is the genuinely new case.

Conjecture 3.18 (Graded algebras of polynomial growth)**.** *Let $\mathfrak{L} = \bigoplus_{m\in\mathbb{Z}} \mathfrak{L}_m$ be a $\mathbb{Z}$-graded Lie algebra over k with $\dim_{\mathsf{k}} \mathfrak{L}_m < \infty$ for all m and $\dim_{\mathsf{k}} \mathfrak{L} = \infty$. Assume that the growth of the homogeneous components is polynomial, i.e. there are $C > 0$ and $q \geq 0$ with $\sum_{|m|\leq r} \dim_{\mathsf{k}} \mathfrak{L}_m \leq Cr^q$ for all $r \geq 1$, and that $\mathfrak{L}$ is graded simple. Then $\mathcal{U}(\mathfrak{L})$ is not noetherian. More generally, the conclusion is claimed in the literature for graded algebras in that class, and only there.*[3]

Remark 3.19 (Why this is a conjecture here; the missing step: the degree-by-degree growth control for the truncated chain). *(1) Grading and finiteness.* By hypothesis $\mathfrak{L} = \bigoplus_{m\in\mathbb{Z}} \mathfrak{L}_m$ with $[\mathfrak{L}_m, \mathfrak{L}_{m'}] \subseteq \mathfrak{L}_{m+m'}$ and $\dim_{\mathsf{k}} \mathfrak{L}_m < \infty$ for every m. Every element of $\mathfrak{L}$ therefore lies in a finite sum of finitely many homogeneous components, each of which is a finite-dimensional k-vector space; this is a restatement of the hypotheses, not an independent result of this note.

(2) The growth control (the step not reconstructed here). The polynomial-growth hypothesis $\sum_{|m|\leq r} \dim_{\mathsf{k}} \mathfrak{L}_m \leq Cr^q$ is what replaces the uniform boundedness of the homogeneous components used in Conjecture 3.13: for the truncated chain to be shown strict one must compare the graded components degree by degree against the growth function. That comparison is not carried out in this note.

(3) Conclusion, conditionally on the imported direction. Granting the published argument for graded algebras of polynomial growth, (1) and (2) put $\mathfrak{L}$ inside its hypotheses and $\mathcal{U}(\mathfrak{L})$ is not

[3]Reported, not reconstructed: the polynomial-growth hypothesis replaces the uniform boundedness of the homogeneous components used in Conjecture 3.13, and the argument for the strictness of the truncated chain must then control the growth degree by degree. This note does not reproduce that control, so the statement is asserted under the hypotheses as stated and no others.

noetherian. The step is an import from the literature, not a proof produced here, and the statement holds only under the hypotheses written above and under no others.

Conjecture 3.20 (Finitely generated infinite-step solvable algebras)**.** *Let $N \geq 1$, let S_N denote the class of finitely generated infinite-dimensional Lie algebras $\mathfrak{g}$ over k that are solvable of derived length N, i.e. $D_N\mathfrak{g} = 0 \neq D_{N-1}\mathfrak{g}$ for the derived series $D_0\mathfrak{g} = \mathfrak{g}$, $D_{k+1}\mathfrak{g} = [D_k\mathfrak{g}, D_k\mathfrak{g}]$. Then, in the direction of the known results, for every $\mathfrak{g} \in S_N$ with $N \geq 1$ the enveloping algebra $\mathcal{U}(\mathfrak{g})$ is not noetherian, with one exception: in the class S_2 there are algebras whose enveloping algebra fails to be noetherian while being a Krull domain, so that non-noetherianity there cannot be detected by the failure of being a domain or of being a Krull domain.*[4]

Remark 3.21 (Why this is a conjecture here; the missing step: an explicit strictly ascending chain of two-sided ideals, with its strictness certified inside each derived layer). *(1) Reading the class.* By definition $D_0\mathfrak{g} = \mathfrak{g}$ and $D_{k+1}\mathfrak{g} = [D_k\mathfrak{g}, D_k\mathfrak{g}]$, so $\mathfrak{g} \in S_N$ means: $\mathfrak{g}$ is finitely generated, $\dim_{\mathsf{k}} \mathfrak{g} = \infty$, and $D_N\mathfrak{g} = 0 \neq D_{N-1}\mathfrak{g}$. Thus $\mathfrak{g}$ has a finite generating set $x_1, \ldots, x_r$ and a finite derived series of length N; both facts are hypotheses here, not consequences derived in this note.

(2) The chain step (imported, not reconstructed). What has to be shown for a member of S_N is the existence of a strictly ascending chain of two-sided ideals of $\mathcal{U}(\mathfrak{g})$. Since $\mathfrak{g}$ is infinite dimensional while $D_N\mathfrak{g} = 0$, one filters $\mathfrak{g}$ by the ideals of the derived series and passes to the induced filtration of $\mathcal{U}(\mathfrak{g})$; the strictness of the resulting chain must be verified inside each derived layer using finite generation of $\mathfrak{g}$ and the non-vanishing of the last nonzero derived ideal $D_{N-1}\mathfrak{g}$. That verification is the step of the published argument, and it is *not* carried out here: no chain is written down explicitly in this note, and no rank computation certifying its strictness is performed. Consequently the conclusion below is a conditional one, valid only in the direction established in the literature under the hypotheses of the statement.

(3) Conclusion, granted (2). If such a strictly ascending chain of two-sided ideals exists, then $\mathcal{U}(\mathfrak{g})$ fails the ascending chain condition on two-sided ideals, hence fails to be left noetherian (a left noetherian ring satisfies the ascending chain condition on two-sided ideals as well). The parenthetical clause about S_2 and Krull domains is likewise a report from the literature and is not checked here; it is recorded only to delimit which ring-theoretic properties can serve as a test for noetherianity in this class.

[4] Reported, not reconstructed. The statement is asserted only for the class S_N as defined (finitely generated, infinite dimensional, solvable of derived length N) and says nothing about solvable algebras that are not finitely generated, nor about the converse direction. In particular the "S_2 but Krull" clause is a report from the literature and is not verified here; it is included because it delimits which ring-theoretic properties can be used as a test for noetherianity in the solvable class.

4 Discussion and open problems

The positive direction is settled (Theorem 3.4), the abelian case is settled (Theorem 3.6), and the graded obstruction covers the graded simple families with finite-dimensional homogeneous components (Conjecture 3.13 and its instances). What remains is exactly the middle ground.

Conjecture 4.1 (Sierra–Walton, as formulated in [SW14])**.** *Let* k *be a field of characteristic* 0 *and let* $\mathfrak{g}$ *be a Lie algebra over* k*. If* $\dim_{\mathsf{k}} \mathfrak{g} = \infty$*, then* $\mathcal{U}(\mathfrak{g})$ *is not noetherian. Equivalently:* $\mathcal{U}(\mathfrak{g})$ *noetherian implies* $\dim_{\mathsf{k}} \mathfrak{g} < \infty$*.*

Open problem 4.2 (What is missing)**.** Conjecture 4.1 is open. Concretely, the following are missing, and each is a separate task:

(1) *A construction.* No example is known of an infinite-dimensional non-abelian $\mathfrak{g}$ with $\mathcal{U}(\mathfrak{g})$ noetherian, and no example is known of an infinite-dimensional $\mathfrak{g}$ with $\mathcal{U}(\mathfrak{g})$ noetherian other than in the abelian (hence non-noetherian) direction. What is missing is either such a construction, or a theorem excluding it under stated extra hypotheses.

(2) *A converse under additional hypotheses.* There is no theorem of the form "if $\mathfrak{g}$ is infinite dimensional and nilpotent (resp. solvable, resp. of bounded growth, resp. graded in some class other than the graded-simple ones above), then $\mathcal{U}(\mathfrak{g})$ is not noetherian", except in the classes listed in Conjectures 3.16, 3.18 and 3.20, each of which holds only under its own hypotheses.

(3) *A transfer principle for finite-codimension subalgebras.* The step "$\mathcal{U}(\mathfrak{h})$ noetherian $\Leftrightarrow$ $\mathcal{U}(\mathfrak{g})$ noetherian for $\dim_{\mathsf{k}}(\mathfrak{g}/\mathfrak{h}) < \infty$" is used for individual pairs in the literature, but Proposition 3.9 shows that the naive route through module finiteness is not available in general (Example 3.10); a proof of the transfer principle, or a counterexample to it, is not in this note.

(4) *Independence of the classes.* The graded obstruction is stated per graded-simple family; it is not known here how far the several families overlap, and no claim is made that the list in Section 3 is exhaustive.

Remark 4.3 (On the status of the statements). Items (1)–(4) are the honest gaps. The results of Section 3 that are proved here (Theorem 3.4, Theorem 3.6, Lemma 3.1, Lemma 3.2, Lemma 3.3, Lemma 3.7, Theorem 3.8, Proposition 3.9, Example 3.10) are proved under exactly the hypotheses stated, in characteristic 0 where the anti-automorphism argument is used. The results reported rather than proved (Conjecture 3.13 and its instances 3.15, 3.16, 3.18, 3.20) are asserted only under their published hypotheses and are marked in place; the Witt case (Corollary 3.15) rests on the standard structure theory of W, which is also not reconstructed here.

参考文献

[KN87] I. M. Krichever and S. P. Novikov, *Algebras of Virasoro type, Riemann surfaces and structures of the theory of solitons*, Funktsional. Anal. i Prilozhen. **21** (1987), no. 2, 46–63.

[MR01] J. C. McConnell and J. C. Robson, *Noncommutative Noetherian Rings*, Graduate Studies in Mathematics **30**, American Mathematical Society, Providence, RI, 2001.

[SW14] S. Sierra and C. Walton, *The universal enveloping algebra of the Witt algebra is not noetherian*, Adv. Math. **262** (2014), 239–260.

A Computational evidence

This appendix collects all computational evidence referred to anywhere in this paper: the two finite computations (A) and (B), the algorithm used for each, its parameter domain, the resulting quantities, and the internal record identifiers together with how to rerun them. Nothing here enters the proofs of Section 3; every conclusion that rests on these computations depends on that evidence and on the stated bounds, and no computation below is a proof of anything.

A.1 Computation (A): filtered dimensions in dimension three. *Quantity computed.* The dimension of the n-th filtered piece $\mathcal{U}_n(\mathfrak{g})$ for $\dim_{\mathsf{k}} \mathfrak{g} = 3$ and $n = 0, 1, \ldots, 8$, and the third finite difference of that sequence. *Algorithm.* Ordered monomials in a basis of $\mathfrak{g}$ are multiplied modulo the span of the monomials of lower total degree, and the dimension is read off as a rank over k; this is a direct comparison with the PBW basis of Lemma 3.1. *Parameter domain.* $d = 3$, with the abelian, Heisenberg and $\mathfrak{sl}_2$ brackets, and $0 \le n \le 8$; no other dimension and no larger n was run.

n	$\dim \mathcal{U}_n$ (all three algebras)	third finite difference
0	1	–
1	4	–
2	10	–
3	20	1
4	35	1
5	56	1
6	84	1
7	120	1
8	165	1

表 1: Computation (A): the computed dimension agrees with $\binom{n+3}{3}$ for $n \le 8$, and the third finite difference is 1 on the computed range. Bound of the computation: $\dim \mathfrak{g} = 3$, $n \le 8$.

A.2 Computation (B): strictness of a truncated family of ideals. *Quantity computed.* For $n = 1, \ldots, 12$ the inclusion $I_n \subsetneq I_{n+1}$ for $I_n = (x_1, \ldots, x_n)$, inside a polynomial ring in countably many indeterminates truncated at degree 12. *Algorithm.* For each n a separating functional on the degree-$(n{+}1)$ part is exhibited exactly: the algebra map killing $x_1, \ldots, x_n$ and fixing x_{n+1} vanishes on I_n and is nonzero on x_{n+1}. The degree truncation enters only the bookkeeping of which monomials were enumerated, not the certificate. *Parameter domain.* $1 \le n \le 12$, degree bound 12; no larger n was run.

family index n	degree bound	enumerated monomials (order)	inclusion strict?
1–12	12	10^2–10^3	yes, certified exactly

表 2: Computation (B): for each $n \le 12$ the inclusion $I_n \subsetneq I_{n+1}$ is certified by a separating functional on the degree-$(n{+}1)$ part. Bound: $n \le 12$.

Computations.

- `exp:000011` — filtered dimension count, $\dim_{\mathsf{k}} \mathfrak{g} = 3$ (abelian, Heisenberg, $\mathfrak{sl}_2$), $n = 0, \ldots, 8$; parameter set $\{d = 3,\ n_{\max} = 8\}$; script digest recorded with the experiment (`e435e239ba21f72f`); conclusion recorded as `supports` with the explicit scope note "$d = 3$ only; does not address noetherianity". Rerun: same script, same parameters.
- `exp:000012` — truncated family of ideals I_n, $n = 1, \ldots, 12$, members $I_1, \ldots, I_{12}$, exact separation test on the degree-$(n{+}1)$ part; parameter set $\{n_{\max} = 12\}$; script digest recorded with the experiment (`3473d47b6f460c3a`); conclusion recorded as `supports`.
- Both records have `method=numeric` and are the source of the two tables in Appendix A. Rerunning either with a different $n_{\max}$ or a different d requires a new record; the present tables must not be extrapolated.

Statements used.

- `stmt:000007` — finite dimension implies noetherian (statement level, no proof text locally).
- `stmt:000004` — infinite-dimensional abelian implies not noetherian.
- `stmt:000005` — finite-codimension transfer, in the form discussed in Proposition 3.9.
- `stmt:000010` — graded-simple infinite-dimensional implies not noetherian (literature direction), the source of Conjecture 3.13.
- `stmt:000018`, `stmt:000019` — derived series and polynomial growth definitions used in Conjectures 3.20 and 3.18.

- `stmt:000001`, `stmt:000002` — Krichever–Novikov algebras and the two conventions for the Witt algebra (used in Definition 2.4).
- `stmt:000003` — the graded-simple definition and the finite-dimensional-homogeneous-components convention.

Literature items.

- `lit:000001` — source for the Krichever–Novikov and Witt definitions (quotation verified verbatim at the pages recorded there).
- `lit:000002` — source for the graded-simple setting and the two Witt conventions (quotation verified verbatim).
- `lit:000004` — McConnell–Robson, *Noncommutative Noetherian Rings*: bibliographic record only, no local full text, so nothing in this note is quoted from it; the reference to Corollary 1.7.4 for the positive direction is recorded at `stmt:000012`.
- `lit:000024` — source for the derived-series and polynomial-growth definitions (quotation verified verbatim).
- `lit:000025` — bibliographic record only (no local full text); it is *not* a source of any verbatim quotation in this note.

How to reproduce the two computations. Both are deterministic scripts run in the workspace; each rerun must pass the parameters named above and must be compared against the digests recorded with `exp:000011` and `exp:000012`. A rerun with a different parameter set produces a different record and does not extend the bounds stated in Appendix A.